

CLIMATESCOPE PROJECT

MILESTONE 1 : DATA PREPARATION AND INITIAL DATA ANALYSIS

In []:

IMPORT REQUIRED LIBRARIES

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from scipy.stats import pearsonr
```

DATA ACQUISITION – LOAD THE DATASET

```
In [2]: data = pd.read_csv("../Dataset/GlobalWeatherRepository.csv")

data.head()
```

Out[2]:

	country	location_name	latitude	longitude	timezone	last_updated_epoch	las
0	Afghanistan	Kabul	34.52	69.18	Asia/Kabul	1715849100	2
1	Albania	Tirana	41.33	19.82	Europe/Tirane	1715849100	2
2	Algeria	Algiers	36.76	3.05	Africa/Algiers	1715849100	2
3	Andorra	Andorra La Vella	42.50	1.52	Europe/Andorra	1715849100	2
4	Angola	Luanda	-8.84	13.23	Africa/Luanda	1715849100	2

5 rows × 41 columns

DATASET STRUCTURE INSPECTION

```
In [3]: data.columns
```

```
Out[3]: Index(['country', 'location_name', 'latitude', 'longitude', 'timezone',
              'last_updated_epoch', 'last_updated', 'temperature_celsius',
              'temperature_fahrenheit', 'condition_text', 'wind_mph', 'wind_kph',
              'wind_degree', 'wind_direction', 'pressure_mb', 'pressure_in',
              'precip_mm', 'precip_in', 'humidity', 'cloud', 'feels_like_celsius',
              'feels_like_fahrenheit', 'visibility_km', 'visibility_miles',
              'uv_index', 'gust_mph', 'gust_kph', 'air_quality_Carbon_Monoxide',
              'air_quality_Ozone', 'air_quality_Nitrogen_dioxide',
              'air_quality_Sulphur_dioxide', 'air_quality_PM2.5', 'air_quality_PM10',
              'air_quality_us-epa-index', 'air_quality_gb-defra-index', 'sunrise',
              'sunset', 'moonrise', 'moonset', 'moon_phase', 'moon_illumination'],
              dtype='object')
```

```
In [4]: data.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 125111 entries, 0 to 125110
Data columns (total 41 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   country                               125111 non-null  object
1   location_name                         125111 non-null  object
2   latitude                             125111 non-null  float64
3   longitude                             125111 non-null  float64
4   timezone                             125111 non-null  object
5   last_updated_epoch                   125111 non-null  int64
6   last_updated                         125111 non-null  object
7   temperature_celsius                  125111 non-null  float64
8   temperature_fahrenheit                125111 non-null  float64
9   condition_text                       125111 non-null  object
10  wind_mph                             125111 non-null  float64
11  wind_kph                             125111 non-null  float64
12  wind_degree                           125111 non-null  int64
13  wind_direction                       125111 non-null  object
14  pressure_mb                           125111 non-null  float64
15  pressure_in                           125111 non-null  float64
16  precip_mm                             125111 non-null  float64
17  precip_in                             125111 non-null  float64
18  humidity                              125111 non-null  int64
19  cloud                                125111 non-null  int64
20  feels_like_celsius                    125111 non-null  float64
21  feels_like_fahrenheit                  125111 non-null  float64
22  visibility_km                         125111 non-null  float64
23  visibility_miles                      125111 non-null  float64
24  uv_index                             125111 non-null  float64
25  gust_mph                             125111 non-null  float64
26  gust_kph                             125111 non-null  float64
27  air_quality_Carbon_Monoxide            125111 non-null  float64
28  air_quality_Ozone                      125111 non-null  float64
29  air_quality_Nitrogen_dioxide            125111 non-null  float64
30  air_quality_Sulphur_dioxide            125111 non-null  float64
31  air_quality_PM2.5                      125111 non-null  float64
32  air_quality_PM10                       125111 non-null  float64
33  air_quality_us-epa-index                125111 non-null  int64
34  air_quality_gb-defra-index              125111 non-null  int64
35  sunrise                                125111 non-null  object
36  sunset                                  125111 non-null  object
37  moonrise                               125111 non-null  object
38  moonset                                125111 non-null  object
39  moon_phase                             125111 non-null  object
40  moon_illumination                      125111 non-null  int64
dtypes: float64(23), int64(7), object(11)
memory usage: 39.1+ MB

```

```
In [5]: data.describe()
```

Out[5]:

	latitude	longitude	last_updated_epoch	temperature_celsius	temperature
count	125111.000000	125111.000000	1.251110e+05	125111.000000	125111.000000
mean	19.193775	21.992531	1.743688e+09	21.590124	21.590124
std	24.425628	65.794125	1.606001e+07	9.599724	9.599724
min	-41.300000	-175.200000	1.715849e+09	-29.800000	-29.800000
25%	3.750000	-6.836100	1.729848e+09	16.300000	16.300000
50%	17.250000	23.240000	1.743671e+09	24.100000	24.100000
75%	40.400000	50.580000	1.757578e+09	28.100000	28.100000
max	64.150000	179.220000	1.771484e+09	49.200000	49.200000

8 rows × 30 columns

In [6]: `data.shape`

Out[6]: (125111, 41)

```
In [7]: print("\n----- NUMERICAL FEATURES -----")
print(data.select_dtypes(include=['int64', 'float64']).columns)

print("\n----- CATEGORICAL FEATURES -----")
print(data.select_dtypes(include=['object']).columns)

----- NUMERICAL FEATURES -----
Index(['latitude', 'longitude', 'last_updated_epoch', 'temperature_celsius',
      'temperature_fahrenheit', 'wind_mph', 'wind_kph', 'wind_degree',
      'pressure_mb', 'pressure_in', 'precip_mm', 'precip_in', 'humidity',
      'cloud', 'feels_like_celsius', 'feels_like_fahrenheit', 'visibility_km',
      'visibility_miles', 'uv_index', 'gust_mph', 'gust_kph',
      'air_quality_Carbon_Monoxide', 'air_quality_Ozone',
      'air_quality_Nitrogen_dioxide', 'air_quality_Sulphur_dioxide',
      'air_quality_PM2.5', 'air_quality_PM10', 'air_quality_us-epa-index',
      'air_quality_gb-defra-index', 'moon_illumination'],
      dtype='object')

----- CATEGORICAL FEATURES -----
Index(['country', 'location_name', 'timezone', 'last_updated',
      'condition_text', 'wind_direction', 'sunrise', 'sunset', 'moonrise',
      'moonset', 'moon_phase'],
      dtype='object')
```

MISSING VALUE ANALYSIS

In [8]: `data.isnull().sum()`

```
Out[8]: country                                0
location_name                                0
latitude                                    0
longitude                                    0
timezone                                    0
last_updated_epoch                          0
last_updated                                0
temperature_celsius                         0
temperature_fahrenheit                     0
condition_text                              0
wind_mph                                    0
wind_kph                                    0
wind_degree                                 0
wind_direction                              0
pressure_mb                                 0
pressure_in                                 0
precip_mm                                   0
precip_in                                   0
humidity                                    0
cloud                                        0
feels_like_celsius                          0
feels_like_fahrenheit                       0
visibility_km                               0
visibility_miles                            0
uv_index                                    0
gust_mph                                    0
gust_kph                                    0
air_quality_Carbon_Monoxide                 0
air_quality_Ozone                           0
air_quality_Nitrogen_dioxide                 0
air_quality_Sulphur_dioxide                 0
air_quality_PM2.5                           0
air_quality_PM10                             0
air_quality_us-epa-index                     0
air_quality_gb-defra-index                   0
sunrise                                      0
sunset                                       0
moonrise                                     0
moonset                                     0
moon_phase                                   0
moon_illumination                           0
dtype: int64
```

Checking Duplicates

```
In [9]: data.duplicated().sum()
```

```
Out[9]: np.int64(0)
```

Convert Units

```
In [10]: # Convert Fahrenheit to Celsius (example column)
data["temp_c_from_f"] = (data["temperature_fahrenheit"] - 32) * 5/9
```

```
data[["temperature_fahrenheit", "temp_c_from_f"]].head()
```

```
Out[10]:
```

	temperature_fahrenheit	temp_c_from_f
0	79.8	26.555556
1	66.2	19.000000
2	73.4	23.000000
3	43.3	6.277778
4	78.8	26.000000

DATA NORMALIZATION

```
In [11]: from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
numeric_columns = data.select_dtypes(include=['int64', 'float64']).columns
data[numeric_columns] = scaler.fit_transform(data[numeric_columns])

print("\nNormalization completed")
```

Normalization completed

Aggregate Data (Daily → Monthly Average)

```
In [12]: data["last_updated"] = pd.to_datetime(data["last_updated"])

monthly_data = data.groupby(
    data["last_updated"].dt.to_period("M")
).mean(numeric_only=True)

monthly_data = monthly_data.reset_index()

monthly_data.head()
```

```
Out[12]:
```

	last_updated	latitude	longitude	last_updated_epoch	temperature_celsius	temperature
0	2024-05	0.574965	0.554293	0.011268	0.695610	
1	2024-06	0.573443	0.555801	0.047542	0.712102	
2	2024-07	0.572694	0.556955	0.094316	0.716461	
3	2024-08	0.572853	0.556652	0.143049	0.716325	
4	2024-09	0.572446	0.556818	0.190377	0.695246	

5 rows × 32 columns

Save the Aggregated Data

```
In [13]: monthly_data.to_csv("../outputs/monthly_weather_data.csv", index=False)
```

Cleaned dataset exported

```
In [14]: data.to_csv("../outputs/cleaned_weather_data.csv", index=False)

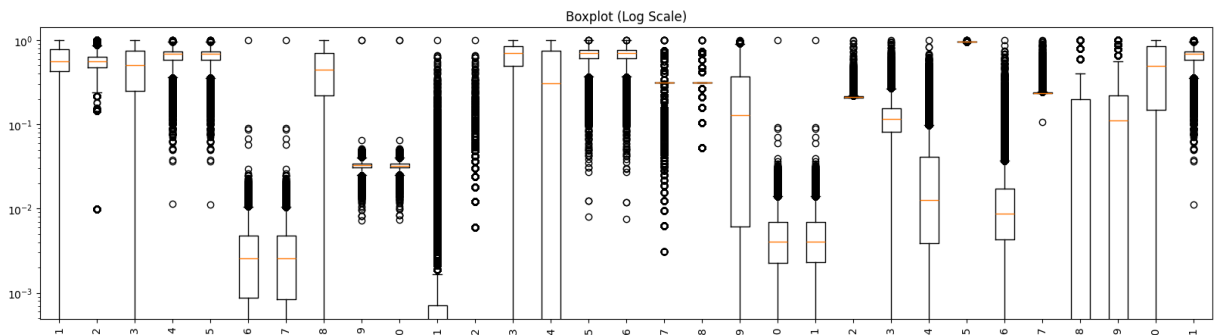
monthly_data.to_csv("../outputs/monthly_weather_data.csv", index=False)

print("\nCleaned dataset exported successfully")
```

Cleaned dataset exported successfully

Boxplot

```
In [15]: plt.figure(figsize=(20, 5))
plt.boxplot(data.select_dtypes(include='number'))
plt.yscale('log')
plt.xticks(rotation=90)
plt.title("Boxplot (Log Scale)")
plt.show()
```



```
In [ ]:
```